

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AC1-021***

Yukon-Hatfield 500 kV

February 2017

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Tenaska Pennsylvania Partners LLC (“Interconnection Customer”) has proposed ***an incremental increase (uprate)*** to prior queue requests T174 and AB1-106 consisting of natural gas, 2x1 combined cycle generating facility in Westmoreland County, PA. The increased capability associated with queue position AC1-106 is achieved through planned efficiencies that maximize the utilization of existing plant equipment to increase its output. The following table shows the values representing the incremental MFO MW for each unit:

	Queue No.	CT1 Unit	CT2 Unit	ST Unit	Total MW
	T174	288.2	288.2	353.6	930
	AB1-106	0	0	10	10
	AC1-021	55	55	0	110
Total MFO MW		343.2	343.2	363.6	1050

The installed facilities will have a total capability of 1050 MW (MFO) with 940 MW of this output being recognized by PJM as Capacity Interconnection Rights (CIR). The proposed in-service date for this project is January 1, 2019. **This study does not imply a West Penn Power Company (“Transmission Owner”) commitment to this in-service date.**

Point of Interconnection

AC1-021 project will deliver power to the West Penn transmission system using the same Point of Interconnection (POI) of prior queue requests T174 and AB1-106. This POI is inside the

Rhodes Lane substation as shown in the one-line diagram in Appendix 2. Rhodes Lane substation is a three breaker ring bus station that is located approximately 2.8 miles away from Yukon substation and approximately 29.6 miles from Hatfield substation. The combined cycle generation site is located approximately 1.5 miles from Rhodes Lane switching station.

Cost Summary

There are no costs associated with the AC1-021 project.

Please Note: All network upgrades and costs associated with the Yukon-Hatfield 500 kV project are shown in the Facilities Study Reports of prior queue requests (T174 and AB1-106) which are published on PJM website at http://www.pjm.com/pub/planning/project-queues/facilities/t174_fac.pdf for T174 and at http://www.pjm.com/pub/planning/project-queues/facilities/ab1106_fac.pdf for AB1-106.

Interconnection Customer Requirements

None.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

FirstEnergy Requirements

The Interconnection Customer will be required to comply with all FE Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the "FirstEnergy Requirements for Transmission Connected Facilities" document located at the following links:

<http://www.firstenergycorp.com/feconnect>

<http://www.pjm.com/planning/design-engineering/to-tech-standards.aspx>

Network Impacts

The Queue Project AB1-106 was evaluated as a 40.0 MW (Capacity 40.0 MW) injection into the T-174 500 kV substation (which is a tap of the Hatfield – Yukon 500 kV line) in the APS area. Project AB1-106 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AB1-106 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2019

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

(Results of the steady-state voltage studies should be inserted here)

None

Short Circuit

(Summary of impacted circuit breakers)

None

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission

Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

Not Applicable

Light Load Analysis - 2019

Not Required

System Reinforcements

Short Circuit: None

Stability and Reactive Power Requirement: no mitigations required however, operational mitigations are proposed for maintenance outages. Please refer to Appendix 3 for Stability Study details.

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Light Load Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

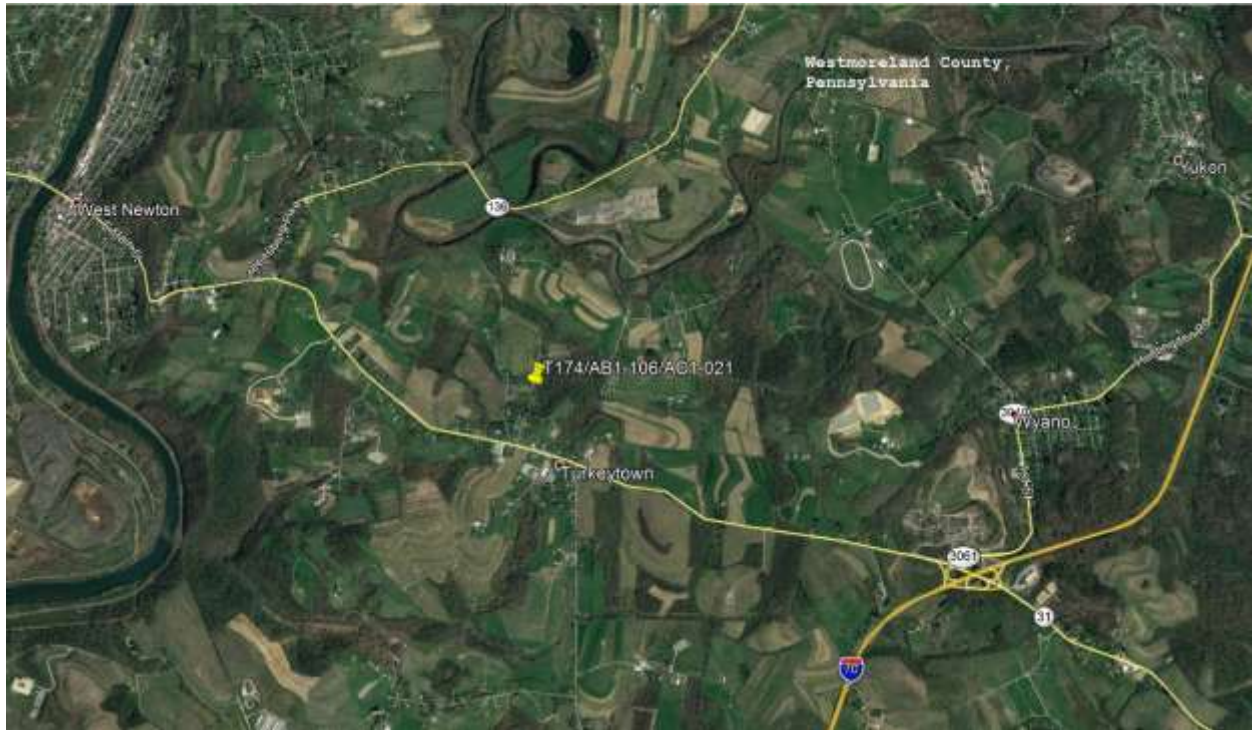
(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

Not Applicable

Appendix 1

Facility Location

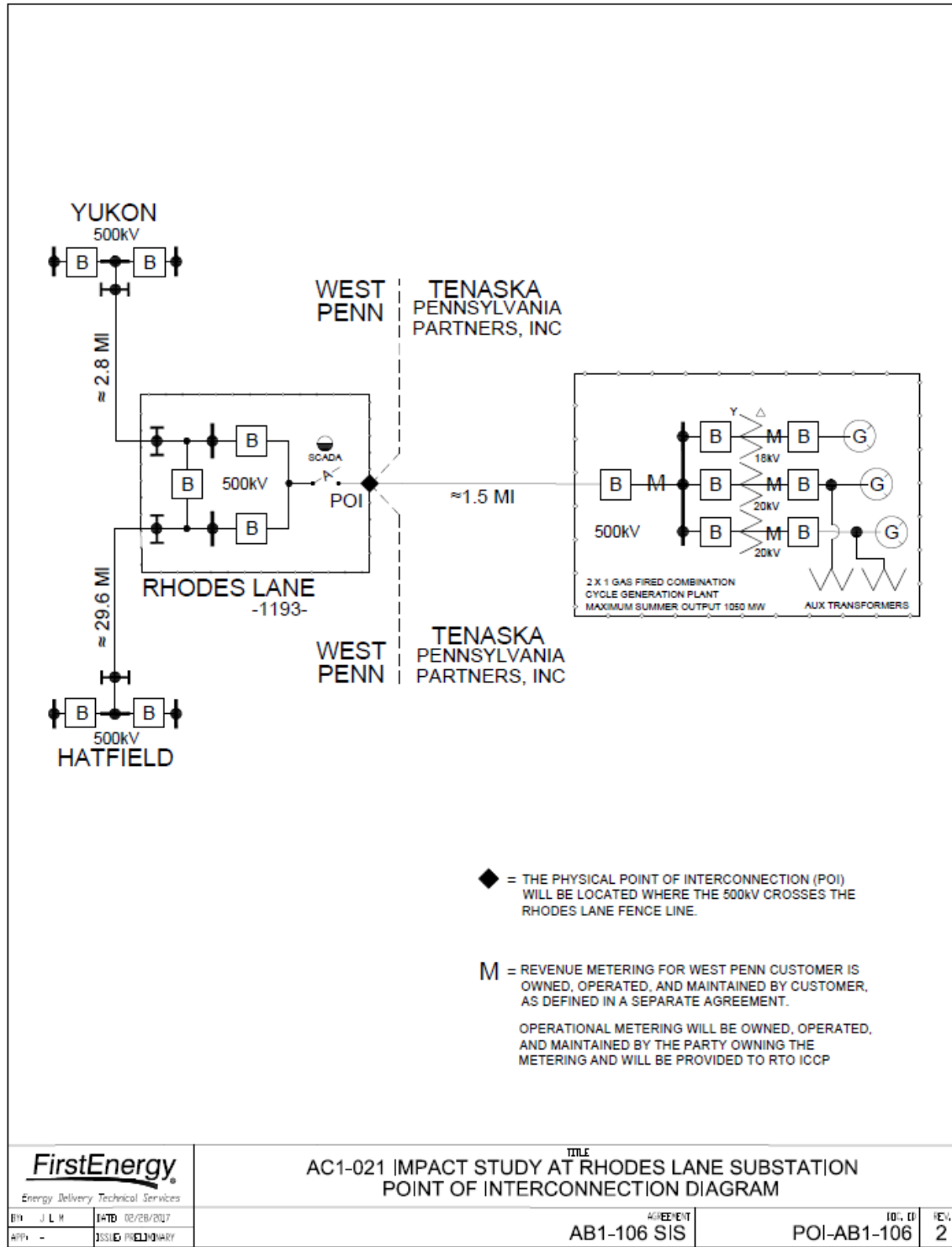
PJM Queue Position: AC1-021



Appendix 2

Interconnection One-Line Diagram

PJM Queue Position: AC1-021



Appendix 3

Dynamic Simulation Analysis (Stability Study)

PJM Queue Position: AC1-021

AC1-021

System Impact Study

Dynamic Simulation Analysis

Version 0
October 03, 2017

TABLE OF CONTENTS

Executive Summary.....	3
1. Introduction	5
2. Description of Project.....	6
3. Loadflow and Dynamics Case Setup	10
4. Fault Cases.....	11
5. Evaluation Criteria	12
6. Summary of Results	13
Attachment 1. Impact Study Data	23
Attachment 2. APS One Line Diagram	23
Attachment 3. PSS/E Model One Line Diagram.....	23
Attachment 4. AC1-021 PSS/E Dynamic Model	23
Attachment 5. AC1-021 PSS/E Case Dispatch.....	23
Attachment 6. Plots from Dynamic Simulations	23

Executive Summary

Generator Interconnection Request AC1-021 is for an increase in energy injection capability of the 2x1 combined cycle facility interconnection request AB1-106. The uprate increases the Maximum Facility Output (MFO) of the plant from 940 MW to 1050 MW. AC1-021 has a Point of Interconnection (POI) at a tap on the Hatfield – Yukon 500 kV line in the Allegheny Power System (APS), Fayette County, Pennsylvania.

This report describes a dynamic simulation analysis of AC1-021 as part of the overall system impact study.

The load flow scenario for the analysis was based on the RTEP 2020 light load case, modified to include applicable queue projects. AC1-021 has been dispatched online at maximum power output, with leading power factor and less than 1.0 pu voltage at the generator bus.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are covered by P1 and P4 contingencies.

The system was tested for a system intact condition and the fault types listed above. Specific fault descriptions and breaker clearing times used for this study are provided in the result table.

No relevant High Speed Reclosing (HSR) contingencies were identified.

Maintenance outages (P6) contingencies are also tested in this study.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

The stability criteria was not met for any of the four maintenance outage contingencies (with a single Ronco unit offline as per the current operational mitigation for these outages) tested on the RTEP 2020 light load case, with results showing,

- AC1-021 units, AA2-173/AB1-106 units and Ronco Units unstable for the P6 contingency MB.P1.18.
- AC1-021 units, AA2-173/AB1-106 units and Ronco Units unstable for the contingency MB.P4.10.

- AC1-021 units, AA2-173/AB1-106 units and Ronco Units unstable for the contingency MB.P4.11.

New operational mitigation for these outages has been proposed which is to reduce AC1-021 generation dispatch by 110 MW and open Ft.Martin-FL-8 breaker.

For the remaining contingencies tested on the 2020 light load case:

- a) The system with AC1-021 included is transiently stable and post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AC1-021 generators were able to ride through faults (except for faults where protective action trips AC1-021).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

No mitigations were found to be required.

1. Introduction

Generator Interconnection Request AC1-021 is for an increase in energy injection capability of the 2x1 combined cycle facility interconnection request AB1-106. The uprate increases the Maximum Facility Output (MFO) of the plant from 940 MW to 1050 MW. AC1-021 has a Point of Interconnection (POI) at a tap on the Hatfield – Yukon 500 kV line in the Allegheny Power System (APS), Fayette County, Pennsylvania.

This analysis is effectively a screening study to determine whether the addition of AC1-021 will meet the dynamic requirements of the NERC, PJM and Transmission Owner reliability standards.

In this report the AC1-021 project and how it is proposed to be connected to the grid are first described, followed by a description of how the project is modeled in this study. The fault cases are then described and analyzed, and lastly a discussion of the results is provided.

2. Description of Project

Generator Interconnection Request AC1-021 is for an increase in energy injection capability of the 2x1 combined cycle facility interconnection request AB1-106. The uprate increases the Maximum Facility Output (MFO) of the plant from 940 MW to 1050 MW. AC1-021 has a Point of Interconnection (POI) at a tap on the Hatfield – Yukon 500 kV line in the Allegheny Power System (APS), Fayette County, Pennsylvania.

Figure 1 shows the simplified one-line diagram of the AC1-021 loadflow model. Table 1 lists the parameters given in the impact study data and the corresponding parameters of the AC1-021 loadflow model.

The dynamic model for AC1-021 is based on standard PSS/E models that were used in the prior queue project AB1-106.

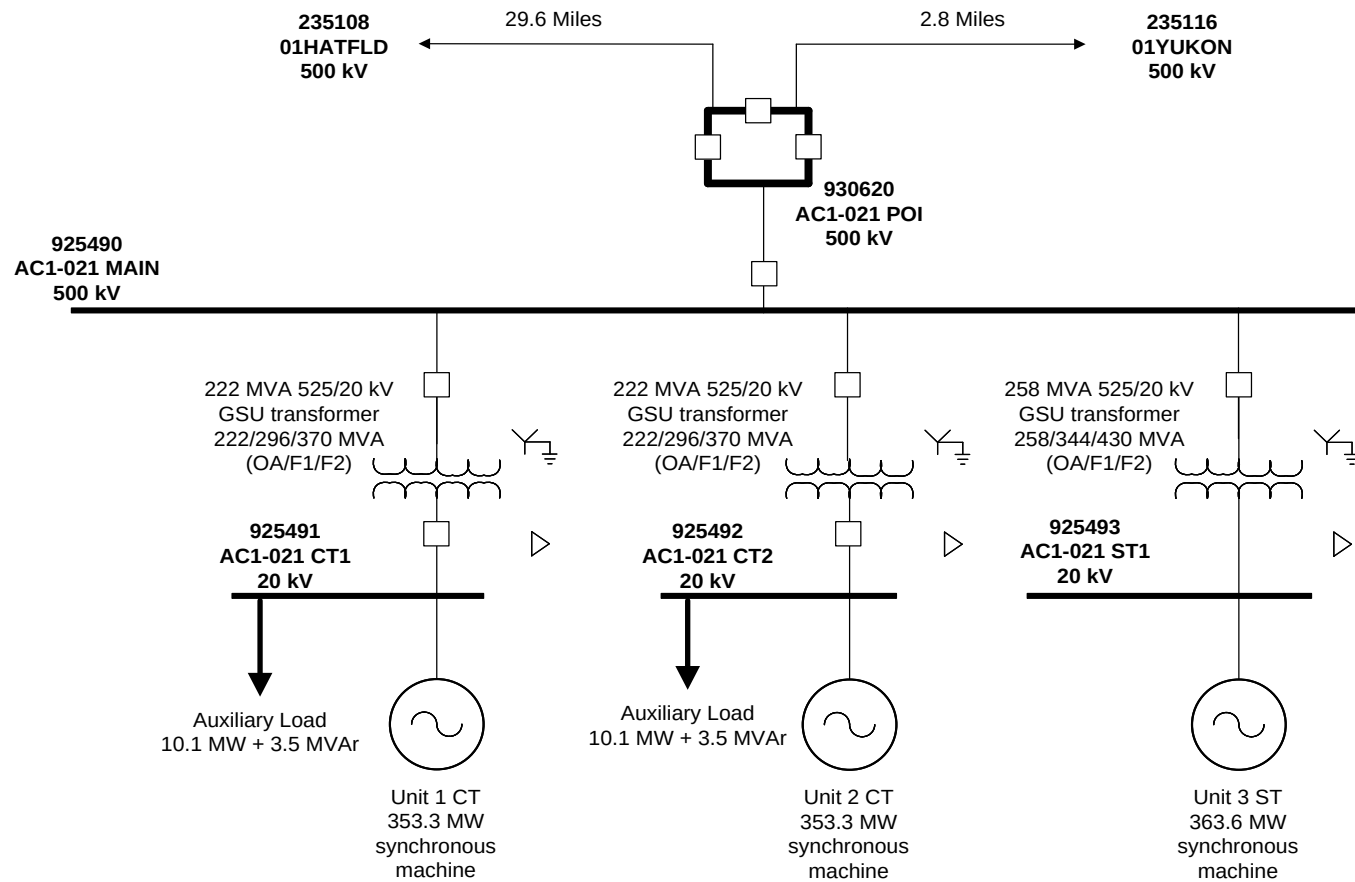


Figure 1: AC1-021 Plant Model

Table 1: AC1-021 Plant Model

	Impact Study Data	Model
Combustion Turbine Generators	2 x 353.3 MW generators MVA base = 435 MVA Vt = 20 kV Unsaturated sub-transient reactance = 0.230 pu @ MVA base	2 x 353.3 MW generators Pgen 353.3 MW Pmax 353.3 MW Pmin 185 MW Qgen 118.5 MVar Qmax 219 MVar Qmin -116 MVar Mbase 435 MVA Zsorce j0.230 pu @ Mbase
Steam Turbine Generator	1 x 363.6 MW generators MVA base = 463 MVA Vt = 20 kV Unsaturated sub-transient reactance = 0.3525 pu @ MVA base	1 x 363.6 MW generators Pgen 363.6 MW Pmax 363.6 MW Pmin 215 MW Qgen 118.5 MVar Qmax 225.3 MVar Qmin -127.0 MVar Mbase 463 MVA Zsorce j0.3525 pu @ Mbase
CT GSU transformers	2x 525/20 kV transformers Rating = 222/296/370 MVA (OA/F1/F2) Transformer base = 222 MVA Impedance = 0.00128 + j0.105 pu @ MVA base Number of taps = NA Tap step size = NA	2x 525/20 kV two winding transformer Rating = 222/296/370 MVA Transformer base = 222 MVA Impedance = 0.00128 + j0.105 pu @ MVA base Number of taps = 5 Tap step size = 2.5 %
ST GSU transformer	1x 525/20 kV transformer Rating = 258/344/430 MVA (OA/F1/F2) Transformer base = 258 MVA Impedance = 0.0015 + j0.105 pu @ MVA base Number of taps = NA Tap step size = NA	1x 525/20 kV two winding transformer Rating = 258/344/430 MVA Transformer base = 258 MVA Impedance = 0.0015 + j0.105 pu @ MVA base Number of taps = 5 Tap step size = 2.5 %
Auxiliary load	11.5 MW + 3.5 MVar at LV side of CT1 GSU 11.5 MW + 3.5 MVar at LV side of CT2 GSU	10.1 MW + 3.5 MVar at LV side of CT1 GSU 10.1 MW + 3.5 MVar at LV side of CT2 GSU
Station Load	N/A	N/A

	Impact Study Data	Model
Transmission line	Impedance=$0.00017 + j 0.00421$ pu; Charging Susceptance = 0.02857 pu @ 100 MVA, 500 kV base. Line length = 1.67 miles	Impedance=$0.00017 + j 0.00421$ pu; Charging Susceptance = 0.02857 pu @ 100 MVA, 500 kV base. Line length = 1.67 miles

3. Loadflow and Dynamics Case Setup

The dynamics simulation analysis was carried out using PSS/E Version 33.7.

The load flow scenario and fault cases for this study are based on PJM's Regional Transmission Planning Process¹.

The selected load flow scenario is the RTEP 2020 light load case with the following modifications:

- a) Addition of all applicable queue projects prior to AC1-021.
- b) Addition of AC1-021 queue project.
- c) Removal of withdrawn and subsequent queue projects in the vicinity of AC1-021.
- d) Dispatch of units in the PJM system to maintain slack generators within limits.

The AC1-021 initial conditions are listed in Table , indicating maximum power output, with unity power factor at the generator bus.

Table 2: AC1-021 machine initial conditions

Bus	Name	Unit	PGEN	QGEN	ETERM	POI Voltage
930622	AC1-021 CT	1	353.3	0 MVar	1.00 pu	1.023 pu
930623	AC1-021 CT	1	353.3	0 MVar	1.00 pu	
930624	AC1-021 ST	1	363.6	0 MVar	1.01 pu	

Generation within the PJM500 system (area 225 in the PSS/E case) and within the vicinity of AC1-021 has been dispatched online at maximum output (P_{MAX}). The dispatch of generation in the vicinity of AC1-021 is given in Attachment 5.

¹ Manual 14B: PJM Region Transmission Planning Process, Rev 33, May 5 2016, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

4. Fault Cases

Tables 3 to 10 list the contingencies that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 10 second simulation time interval.

The studied contingencies include:

- a) Steady state operation (Category P0);
- b) Three-phase (3ph) fault with normal clearing (Category P1)
- c) Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
- d) Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
- e) Three phase faults with normal clearing time with maintenance outages of Fort Martin – Ronco 500 kV line;
- f) Three phase faults with normal clearing time with maintenance outages of AC1-021 – Yukon 500 kV line;
- g) Single phase faults with stuck breaker with maintenance outages of AC1-021 – Yukon 500 kV line; and
- h) Three phase faults with stuck breaker with maintenance outages of AC1-021 – Yukon 500 kV line.

No relevant Bus, Tower or High Speed Reclosing (HSR) contingencies were identified.

Clearing times listed in Tables 3 to 10 are as per Revision 18 of “*2016 Revised Clearing times for each PJM company*” spreadsheet.

Attachment 2 contains the one-line diagrams of the APS network in the vicinity of AC1-021, showing where faults were applied.

The positive sequence fault impedances for single line to ground faults were derived from a separate short circuit case, modified to ensure that connected generators in the vicinity of AC1-021 have not withdrawn from the PJM queue, and are not greater than the queue position under study.

5. Evaluation Criteria

This study is focused on AC1-021, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process and Transmission Owner criteria:

- a) The system with AC1-021 included is transiently stable and post-contingency oscillations should be positively damped with a damping margin of at least 3%.
- b) The AC1-021 is able to ride through faults (except for faults where protective action trips AC1-021).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

6. Summary of Results

Plots from the dynamic simulations are provided in Attachment 6, with results summarized in Table 3 through Table 10.

The stability criteria was not met for any of the four maintenance outage contingencies (with a single Ronco unit offline as per the current operational mitigation for these outages) tested on the RTEP 2020 light load case, with results showing,

The stability criteria was not met for any of the four maintenance outage contingencies (with a single Ronco unit offline as per the current operational mitigation for these outages) tested on the RTEP 2020 light load case, with results showing,

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- AC1-021 units, AA2-173/AB1-106 units and Ronco Units unstable for the contingency MB.P4.10.
- AC1-021 units, AA2-173/AB1-106 units and Ronco Units unstable for the contingency MB.P4.11.

New operational mitigation for these outages has been proposed which is to reduce AC1-021 generation dispatch by 110 MW and open Ft.Martin-FL-8 breaker.

Damping of oscillations seen in the Keystone station generator G1 PELEC was confirmed to be >3%.

For the remaining 30 fault contingencies tested on the 2020 light load case:

- a) The system with AC1-021 included is transiently stable and post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AC1-021 generators were able to ride through faults (except for faults where protective action trips AC1-021).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

7. Mitigations

Following new operational mitigation has been proposed to alleviate the instabilities due to maintenance outages listed in Table 8, 9 and 10.

1. Reduce AC1-021 Generation Dispatch by 110 MW
2. Open the Ft. Martin FL-8 Breaker

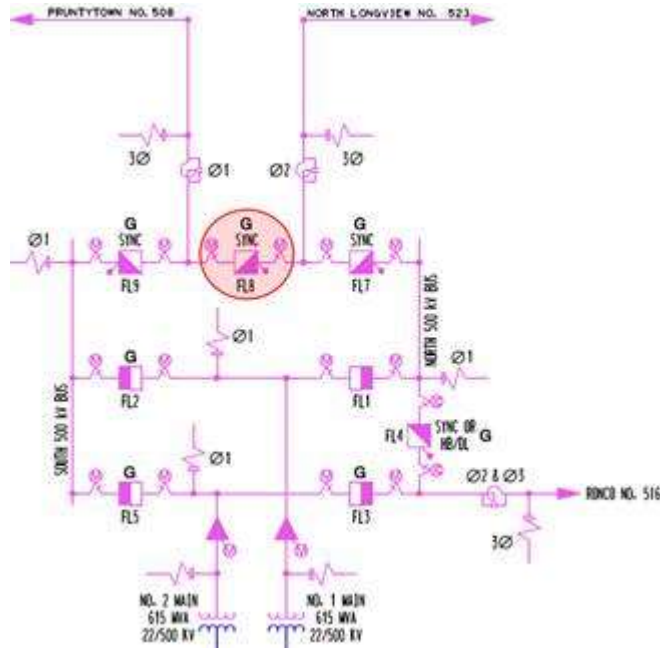


Table 3: Steady State Operation

Fault ID	Duration	AC1-021 No Mitigation
P0.00	Steady state 20 sec	Stable

Table 4: Three-phase Faults With Normal Clearing

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	AC1-021 No Mitigation
P1.01	Fault at AC1-021 500 kV on AC1-021 circuit (Trips AC1-021 CT1, CT2 and ST1).	3	Stable
P1.02	Fault at AC1-021 500 kV on Hatfield circuit HAT-YUK 518.	3	Stable
P1.03	Fault at AC1-021 500 kV on Yukon circuit HAT-YUK 518.	3	Stable
P1.04	Fault at Hatfield 500 kV on AC1-021 circuit HAT-YUK 518.	3	Stable
P1.05	Fault at Hatfield 500 kV on AA2-173 circuit (Trips AA2-173 CT1 and ST1).	3	Stable
P1.06	Fault at Hatfield 500 kV on Ronco circuit HF-RC 538.	3	Stable
P1.07	Fault at Hatfield 500 kV on Black Oak circuit HAT-BO 542.	3	Stable
P1.08	Fault at Yukon 500 kV on AC1-021 circuit HAT-YUK 518	3	Stable
P1.09	Fault at Yukon 500 kV on South Bend circuit SBE-YU 507	3	Stable
P1.10	Fault at Yukon 500 kV on 500/138 kV T1 circuit, normal clear loss of T3	3	Stable
P1.11	Fault at Yukon 500 kV on 500/138 kV T2 circuit, normal clear loss of T4	3	Stable
P1.12	Fault at Yukon 500 kV on 500/138 kV T3 circuit, normal clear loss of T1	3	Stable
P1.13	Fault at Yukon 500 kV on 500/138 kV T4 circuit, normal clear loss of T2	3	Stable
P1.14	Fault at Ronco 500kV on Fort Martin circuit	3	Stable
P1.15	Fault on Ronco 500kV on Hatfield circuit	3	Stable
P1.16	Fault on Ronco 500kV on Fayette circuit (Trips Ronco units)	3	Stable
P1.17	Fault on Fort Martin 500kV on Pruntytown circuit	3	Stable
P1.18	Fault on Fort Martin 500kV on Ronco circuit	3	Stable

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	AC1-021 No Mitigation
P1.19	Fault on Fort Martin 500kV on North Longview circuit	3	Stable
P1.20	Fault on Fort Martin 500kV on Fort Martin unit 1 GSU (Trips Fort Martin unit 1)	3	Stable
P1.21	Fault on Fort Martin 500kV on Fort Martin unit 2 GSU (Trips Fort Martin unit 2)	3	Stable
P1.22	Fault on South Bend 500kV on Keystone circuit	3	Stable
P1.23	Fault on South Bend 500kV on Yukon circuit	3	Stable
P1.24	Fault on South Bend 500kV on South Bend collector bus (Trips South Bend units)	3	Stable
P1.25	Fault on Black Oak 500kV on Hatfield circuit	3	Stable
P1.26	Fault on Black Oak 500kV on Bedington circuit	3	Stable
P1.27	Fault on Black Oak 500kV on 500/138kV TF	3	Stable
P1.28	Fault on Black Oak 500kV on Black Oak SVC circuit (Trips Black Oak SVC and capacitors)	3	Stable

Table 5: Single-phase Faults With Stuck Breaker

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	AC1-021 No Mitigation
P4.01	Fault at AC1-021 500 kV on AC1-021 circuit. Breaker stuck to AC1-021. Fault cleared with loss of Hatfield circuit to AC1-021 circuit and Yukon to AC1-021 circuit (Trips AC1-021 units CT1, CT2 and ST1).	3 / 12	Stable
P4.02	Fault at Hatfield 500 kV on AA2-173 circuit. Breaker stuck to AA2-173. Fault cleared with no additional losses (Trips AA2-173 CT1 and ST1).	3 / 12	Stable
P4.03	Fault at Hatfield 500 kV bus, delayed clear Ronco circuit HF-RC 538 due to Breaker 9 stuck.	3 / 12	Stable
P4.04	Fault at Hatfield 500 kV bus, delayed clear Black Oak circuit HAT-BO 542 due to Breaker 6 stuck.	3 / 12	Stable
P4.05	Fault at Hatfield 500 kV bus, delayed clear on AC1-021 circuit HF-YUK 518 due to Breaker 3 stuck.	3 / 12	Stable
P4.06	Fault at Yukon 500 kV on AC1-021 circuit HAT-YUK 518. Breaker YL1 stuck. Cleared with loss of Yukon 500/138 kV Transformer T1 and Yukon 500/138 kV Transformer T3.	3 / 12	Stable
P4.07	Fault at Yukon 500 kV on South Bend circuit SBE-YU 507. Breaker YL6 stuck. Cleared with loss of Yukon 500/138 kV Transformer T2 and Yukon 500/138 kV Transformer T4.	3 / 12	Stable
P4.08	Fault at Yukon 500 kV on Yukon 500/138 kV Transformer T1, normal clear loss of 500/138kV T3. Breaker YL1 stuck. Fault cleared with loss of AC1-021 circuit HAT-YUK 518.	3 / 12	Stable
P4.09	Fault at Yukon 500 kV on Yukon 500/138 kV Transformer T1, normal clear loss of 500/138kV T3.. Breaker YL4 stuck. Fault cleared with loss of South Bend circuit SBE-YU 507 circuit.	3 / 12	Stable

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	AC1-021 No Mitigation
P4.10	Fault at Fort Martin 500 kV on North Longview circuit. Breaker 8 stuck. Fault cleared with loss of Fort Martin – Pruntytown 500kV circuit	3 / 12.5	Stable
P4.11	3ph fault at Fort Martin 500 kV on North Longview circuit. Breaker 8 stuck. Fault cleared with loss of Fort Martin – Pruntytown 500kV circuit	3 / 7.5	Unstable*
P4.12	Fault on Fort Martin 500kV on Fort Martin unit 2. Breaker 3 stuck. Fault cleared with loss of Fort Martin – Ronco 500kV circuit	3 / 12.5	Stable

***Ronco unit 3 would be unstable for this contingency. And the stability would be withstand if this unit is generating reactive power shown in sensitivity study.**

Table 6: Single-phase Faults With Delayed (Zone 2) Clearing at line end closest to AC1-021 POI

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	AC1-021 No Mitigation
P5.01	Fault at 80% of line from AC1-021 500 kV on Yukon circuit HAT-YUK 518. Delayed clearing at AC1-021.	3 / 21	Stable
P5.02	Fault at 80% of line from AC1-021 500 kV on Hatfield circuit HAT-YUK 518. Delayed clearing at AC1-021.	3 / 21	Stable
P5.03	Fault at 80% of line from Hatfield 500 kV on Ronco circuit HF-RC 538. Delayed clearing at Hatfield.	3 / 21	Stable
P5.04	Fault at 80% of line from Hatfield 500 kV on Black Oak circuit HAT-BO 542. Delayed clearing at Hatfield.	3 / 21	Stable
P5.05	Fault at 80% of line from Yukon 500 kV on South Bend circuit SBE-YU_507. Delayed clearing at Yukon.	3 / 21	Stable
P5.06	Fault at 80% of line from Ronco 500 kV on Fort Martin circuit. Delayed clearing at Ronco.	3 / 21	Stable
P5.07	Fault at 80% of line from South Bend 500 kV on Keystone circuit. Delayed clearing at South Bend.	3 / 21	Stable

Table 7: Three-phase Faults With Normal Clearing – Prior outage of Fort Martin–Ronco 500 kV line

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	AC1-021 No Mitigation
MA.P1.03	Fault at AC1-021 500 kV on AC1-021 – Yukon 500 kV line (with AA2-139 Unit 1 offline*)	3	Stable
MA.P1.03	Fault at AC1-021 500 kV on AC1-021 – Yukon 500 kV line	3	Stable

**Table 8: Three-phase Faults With Normal Clearing – Prior outage of AC1-021 –Yukon 500 kV line
(Breaker 8 at Fort Martin is open to avoid stuck breaker faults)**

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	AC1-021 No Mitigation
MB.P1.17	Fault at Fort Martin 500 kV on Fort Martin – Pruntytown 500 kV line (with AA2-139 Unit 1 offline*)	3	Stable
MB.P1.18*	Fault at Fort Martin 500 kV on Fort Martin – Ronco 500 kV line (with AA2-139 Unit 1 offline*)	3	Unstable, curtailed T-174/AB1-106/AC1-021 by 110MW to be stable
MB.P1.19	Fault at Fort Martin 500 kV on Fort Martin – North Longview 500 kV line (with AA2-139 Unit 1 offline*)	3	Stable
MB.P1.17	Fault at Fort Martin 500 kV on Fort Martin – Pruntytown 500 kV line	3	Stable
MB.P1.18*	Fault at Fort Martin 500 kV on Fort Martin – Ronco 500 kV line	3	Unstable, curtailed T-174/AB1-106/AC1-021 by 110MW to be stable
MB.P1.19	Fault at Fort Martin 500 kV on Fort Martin – North Longview 500 kV line	3	Stable

* Most limiting contingency, Hatfield units are all retired in the case.

* As specified in Section 5: Index and Operating Procedures for PJM RTO Operation by the document “PJM Manual 03: Transmission Operations.pdf”, dated April 1, 2013.

Table 9: Three-phase Faults With Stuck Breaker – Prior outage of AC1-021 –Yukon 500 kV line

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	AC1-021 No Mitigation
MB.P4.11	Fault at Fort Martin 500 kV on North Longview circuit FTM-NLVW_523. Breaker 8 stuck. Fault cleared with loss of Pruntytown circuit PR-FM_508 (with AA2-139 Unit 1 offline*)	3 / 7.5	Unstable*

*Mitigation plan is to open Breaker 8 at Fort Martin 500kV switchyard to avoid this stuck breaker contingency.

Table 10: Single-phase Faults With Stuck Breaker – Prior outage of AC1-021 –Yukon 500 kV line

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	AC1-021 No Mitigation
MB.P4.10	Fault at Fort Martin 500 kV on North Longview circuit FTM-NLVW_523. Breaker 8 stuck. Fault cleared with loss of Pruntytown circuit PR-FM_508 (with AA2-139 Unit 1 offline*)	3 / 12.5	Unstable*

*Mitigation plan is to open Breaker 8 at Fort Martin 500kV switchyard to avoid this stuck breaker contingency.

* As specified in Section 5: Index and Operating Procedures for PJM RTO Operation by the document "PJM Manual 03: Transmission Operations.pdf", dated December 1, 2015.

Attachment 1. Impact Study Data

Attachment 2. APS One Line Diagram

Attachment 3. PSS/E Model One Line Diagram

Attachment 4. AC1-021 PSS/E Dynamic Model

Attachment 5. AC1-021 PSS/E Case Dispatch

Attachment 6. Plots from Dynamic Simulations